

Package ‘cagedr’

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Title Access Novo CAGED Microdata from the Brazilian Ministry of Labour

Version 0.1.0

Description Download and read the public, non-identified microdata of the Novo CAGED (Cadastro Geral de Empregados e Desempregados), the monthly registry of formal employment movements published by the Brazilian Ministry of Labour and Employment through the PDET FTP server [<ftp://ftp.mtps.gov.br/pdet/microdados/>](ftp://ftp.mtps.gov.br/pdet/microdados/). Lists the reference months available on the server, downloads the three monthly files (movements declared on time, declared late, and exclusions) with an idempotent local cache, and reads the national 7z archives as a stream, filtering by state and selecting columns before anything is kept in memory, so that a single state can be extracted without loading the full national file. Also provides the official record layout and a helper to consolidate admissions, separations and net balance by reference month.

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Encoding UTF-8

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URL <https://github.com/StrategicProjects/cagedr>,
<https://strategicprojects.github.io/cagedr/>

BugReports <https://github.com/StrategicProjects/cagedr/issues>

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caged_available	<i>List the reference months published on the PDET/MTE FTP server</i>
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Description

Queries the public FTP server of the Ministry of Labour and Employment and returns the reference months (AAAAMM) of the Novo CAGED that are currently published, with the date each folder was last modified. The Ministry publishes a reference month around the end of the following month, and occasionally re-publishes earlier months.

Usage

caged_available(year = NULL, timeout = 30, verbose = NULL)

Arguments

- | | |
|---------|---|
| year | Optional integer vector of years to restrict the listing (for example 2025:2026). NULL (the default) lists every year since 2020. |
| timeout | Connection timeout in seconds for each FTP request. |
| verbose | Emit progress messages? Defaults to <code>getOption("cagedr.verbose", TRUE)</code> . |

Value

A tibble with one row per reference month and columns period (integer AAAAMM), year, month, modified (POSIXct, folder modification time on the server) and url (the folder URL). Sorted from the most recent to the oldest month. Returns an empty tibble, with a warning, when the server cannot be reached.

See Also

[caged_download\(\)](#) to fetch the files of a month, [caged_periods\(\)](#) to build a sequence of months offline.

Examples

```
# Requires network access to ftp.mtps.gov.br
months <- tryCatch(caged_available(year = 2026), error = function(e) NULL)
if (!is.null(months)) head(months)
```

caged_balance

Consolidate admissions, separations and net balance

Description

Applies the accounting rule of the Novo CAGED to a set of records read with [caged_read\(\)](#) or [caged_fetch\(\)](#) and aggregates them. Movements declared on time (MOV) and late (FOR) count with their own sign; **exclusions** (EXC) cancel a previously declared movement, so their saldomovimentacao is inverted before summing. The aggregation is always by competenciamov, the reference month of the movement, which is what the Ministry's published totals use.

Usage

```
caged_balance(data, by = NULL)
```

Arguments

data	A tibble returned by caged_read() or caged_fetch() . Must contain competenciamov, saldomovimentacao and caged_file.
by	Character vector of additional grouping columns (for example "municipio", "uf", "secao"). competenciamov is always included.

Details

Because FOR and EXC archives of month M carry records of earlier months, the balance of a given month keeps changing as later archives are published. To reproduce the Ministry's figure for a month, read every archive published up to the date of interest.

Value

A tibble with the grouping columns and admissions (records with positive sign after the exclusion rule), separations (negative sign), net (admissions - separations) and records (number of records aggregated). When the data contain salario, also admission_salary (sum of the salaries of the admissions) and mean_admission_salary.

Examples

```
f <- system.file("extdata", "CAGEDMOV202301_sample.7z", package = "cagedr")
x <- caged_read(f, verbose = FALSE)
caged_balance(x, by = "uf")
```

caged_cache_clear	<i>Remove archives from the cache</i>
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Description

Remove archives from the cache

Usage

```
caged_cache_clear(period = NULL, cache_dir = NULL)
```

Arguments

period	Optional reference months (AAAAMM) to remove. NULL removes every cached archive.
cache_dir	Optional path. When given, it is returned as is (after creating the folder).

Value

Invisibly, the number of files removed.

Examples

```
caged_cache_clear()
```

caged_cache_dir	<i>Cache directory used by cagedr</i>
-----------------	---------------------------------------

Description

Downloaded archives are stored in a local cache so that a reference month is never downloaded twice. The location is resolved in this order:

Usage

```
caged_cache_dir(cache_dir = NULL)
```

Arguments

`cache_dir` Optional path. When given, it is returned as is (after creating the folder).

Details

1. the `cache_dir` argument;
2. the `CAGEDR_CACHE_DIR` environment variable;
3. the `cagedr.cache_dir` R option;
4. a session-scoped folder under `tempdir()`, which R removes when the session ends.

Set one of the first three to keep the archives between sessions. Files published by the Ministry are large (about 55 MB per month) and change only when a month is re-published, so a persistent cache is recommended for repeated work.

Value

The cache directory path, created if needed.

Examples

```
caged_cache_dir()

# For a persistent cache, set the environment variable (for instance in
# your .Renvron). Here a temporary folder is used and the previous value
# is restored afterwards.
old <- Sys.getenv("CAGEDR_CACHE_DIR", unset = NA)
Sys.setenv(CAGEDR_CACHE_DIR = file.path(tempdir(), "caged-cache"))
caged_cache_dir()
if (is.na(old)) Sys.unsetenv("CAGEDR_CACHE_DIR") else Sys.setenv(CAGEDR_CACHE_DIR = old)
```

caged_cache_list	<i>List the archives currently in the cache</i>
------------------	---

Description

List the archives currently in the cache

Usage

```
caged_cache_list(cache_dir = NULL)
```

Arguments

cache_dir Optional path. When given, it is returned as is (after creating the folder).

Value

A tibble with columns path, period, file (MOV, FOR or EXC), size_bytes and modified.

Examples

```
caged_cache_list()
```

caged_download	<i>Download the monthly archives of the Novo CAGED</i>
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Description

Fetches the .7z archives of one or more reference months from the PDET/MTE FTP server into the local cache (see [caged_cache_dir\(\)](#)). Archives already in the cache are not downloaded again unless force = TRUE.

Usage

```
caged_download(
  period,
  files = c("MOV", "FOR", "EXC"),
  cache_dir = NULL,
  force = FALSE,
  timeout = 900,
  verbose = NULL
)
```

Arguments

period	Reference months in AAAAMM format (integer or character vector). See caged_periods() and caged_available() .
files	Which files to download: any subset of c("MOV", "FOR", "EXC").
cache_dir	Optional cache directory (see caged_cache_dir()).
force	Re-download archives already in the cache?
timeout	Timeout in seconds for each file. The Ministry's FTP is slow at times; the default allows 15 minutes per file.
verbose	Emit progress messages? Defaults to <code>getOption("cagedr.verbose", TRUE)</code> .

Details

Every reference month has up to three files, all national:

- MOV (CAGEDMOV<AAAAMM>.7z): movements declared on time for that month, the bulk of the data (about 55 MB compressed, 4 to 5 million records);
- FOR (CAGEDFOR<AAAAMM>.7z): movements declared **late**, which refer to earlier reference months (`competenciamov < AAAAMM`);
- EXC (CAGEDEXC<AAAAMM>.7z): **exclusions**, movements cancelled by the employer, also referring to earlier months.

The first months of the series (January to March 2020) have no FOR or EXC files; those are reported as `not_found`, not as errors.

Value

A tibble with one row per (period, file) and columns `period`, `file`, `path` (local path, NA when not available), `status` ("downloaded", "cached", "not_found" or "error") and `url`.

See Also

[caged_read\(\)](#) to read a downloaded archive, [caged_fetch\(\)](#) for download and read in one call.

Examples

```
# Requires network access; downloads about 1 MB (the EXC file is small).
res <- tryCatch(
  caged_download(202401, files = "EXC", cache_dir = tempdir()),
  error = function(e) NULL
)
res
```

caged_fetch

*Download and read Novo CAGED microdata in one call***Description**

Convenience wrapper: `caged_download()` followed by `caged_read()` on every archive found, with the results stacked. Missing files (for example FOR/EXC in early 2020) are skipped with a message.

Usage

```
caged_fetch(
  period,
  uf = NULL,
  files = c("MOV", "FOR", "EXC"),
  columns = NULL,
  cache_dir = NULL,
  force = FALSE,
  types = TRUE,
  chunk_size = 500000L,
  timeout = 900,
  verbose = NULL
)
```

Arguments

period	Reference months in AAAAMM format (integer or character vector). See <code>caged_periods()</code> and <code>caged_available()</code> .
uf	Optional vector of state codes to keep, as IBGE two-digit numbers (for example 26 for Pernambuco, c(26, 25) for Pernambuco and Paraiba). NULL keeps every state.
files	Which files to download: any subset of c("MOV", "FOR", "EXC").
columns	Optional character vector of columns to keep, using the normalized names listed by <code>caged_layout()</code> . NULL keeps all columns. The uf column is always read (it is needed for filtering) but is only returned when requested or when columns is NULL.
cache_dir	Optional cache directory (see <code>caged_cache_dir()</code>).
force	Re-download archives already in the cache?
types	Convert the numeric columns of the layout (codes, salary) to integer/double? If FALSE every column is returned as character.
chunk_size	Number of lines parsed per chunk. Larger chunks are faster but use more memory; the default (500,000 lines) uses well under 1 GB.
timeout	Timeout in seconds for each file. The Ministry's FTP is slow at times; the default allows 15 minutes per file.
verbose	Emit progress messages? Defaults to <code>getOption("cagedr.verbose", TRUE)</code> .

Value

A tibble with the records of every archive read (see [caged_read\(\)](#) for the columns), or an empty tibble when nothing was available. The attribute "download" holds the tibble returned by [caged_download\(\)](#), so that not_found and error files can be inspected.

Examples

```
# Requires network access. Pernambuco, one month, the three files:
pe <- tryCatch(
  caged_fetch(202401, uf = 26, cache_dir = tempdir(), verbose = FALSE),
  error = function(e) NULL
)
if (!is.null(pe)) table(pe$caged_file)
```

caged_layout

Record layout of the Novo CAGED microdata

Description

The columns of the monthly files, with the original (accented) name used by the Ministry, the normalized name returned by [caged_read\(\)](#), the type assigned when types = TRUE, the files in which the column appears and a short description. The official layout (the "Layout Nao-identificado Novo Caged Movimentacao" spreadsheet) is published in the same FTP folder as the data.

Usage

```
caged_layout()
```

Details

Two columns exist only in the exclusions file (EXC): competenciaexc and indicadordeexclusao.

Value

A tibble with columns column (normalized name), original (name in the file), type, files and description.

Examples

```
caged_layout()
subset(caged_layout(), files != "MOV, FOR, EXC")
```

caged_periods	<i>Build a sequence of reference months</i>
---------------	---

Description

Offline helper that expands a range of reference months in AAAAMM format. Useful to request several months from `caged_download()` or `caged_fetch()` without querying the server first.

Usage

```
caged_periods(from, to = from)
```

Arguments

from, to	First and last reference month, as integers or strings in AAAAMM format (for example 202001 and 202612). to defaults to from.
----------	---

Value

An integer vector of reference months, in increasing order.

Examples

```
caged_periods(202301, 202306)
caged_periods("202412")
```

caged_read	<i>Read a Novo CAGED archive as a stream</i>
------------	--

Description

Reads the .txt inside a monthly .7z archive without extracting it to disk and without loading the national file in memory: the text is decompressed as a stream and parsed in chunks, and each chunk is filtered by state and reduced to the requested columns before being kept. This is what makes it practical to extract one state (a few hundred thousand records) from a national file of several million records on a modest machine.

Usage

```
caged_read(
  path,
  uf = NULL,
  columns = NULL,
  types = TRUE,
  chunk_size = 500000L,
  verbose = NULL
)
```

Arguments

path	Path to a CAGEDMOV, CAGEDFOR or CAGEDEXC archive, as returned by caged_download() .
uf	Optional vector of state codes to keep, as IBGE two-digit numbers (for example 26 for Pernambuco, c(26, 25) for Pernambuco and Paraiba). NULL keeps every state.
columns	Optional character vector of columns to keep, using the normalized names listed by caged_layout() . NULL keeps all columns. The uf column is always read (it is needed for filtering) but is only returned when requested or when columns is NULL.
types	Convert the numeric columns of the layout (codes, salary) to integer/double? If FALSE every column is returned as character.
chunk_size	Number of lines parsed per chunk. Larger chunks are faster but use more memory; the default (500,000 lines) uses well under 1 GB.
verbose	Emit progress messages? Defaults to <code>getOption("cagedr.verbose", TRUE)</code> .

Value

A tibble with the selected records and columns, plus two columns added by the package: `caged_file` ("MOV", "FOR" or "EXC", detected from the archive name) and `caged_period` (the reference month of the **archive**, which for FOR and EXC differs from the `competenciamov` of the records). Column names are normalized: accents removed and lower case (`competenciamov`, `municipio`, `salario`). Returns an empty tibble when no record matches.

See Also

[caged_layout\(\)](#) for the meaning of every column, [caged_fetch\(\)](#) for download and read in one call.

Examples

```
# A small sample archive ships with the package (Pernambuco and Bahia rows).
f <- system.file("extdata", "CAGEDMOV202301_sample.7z", package = "cagedr")
x <- caged_read(f, verbose = FALSE)
dim(x)

# One state, a few columns
pe <- caged_read(f, uf = 26,
  columns = c("competenciamov", "municipio", "saldomovimentacao", "salario"),
  verbose = FALSE)

pe
```

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